The Michelson Interferometer

SYNOPSIS

Split a beam in two, send the halves down separate arms, bring them back together and let them interfere. Because the path difference is now **adjustable**, the fringe pattern becomes a ruler — one whose smallest division is a fraction of a wavelength of light.

BEFORE YOU START — PREDICT

You move one mirror by **315 nm**. That is about a three-hundredth of the thickness of a human hair. What would you have to **see** for that movement to be measurable — and could you count it by eye?

PERFORMANCE EXPECTATIONS

1. **Describe** the path of each half-beam through a Michelson interferometer.
2. **Explain** why moving a mirror by $\lambda/2$ shifts the pattern by a whole fringe.
3. **Compute** a mirror displacement from a fringe count, and a fringe count from a displacement.
4. **Apply** the same fringe-counting idea to a **refractive index** measurement with a gas cell.

Michelson interferometer

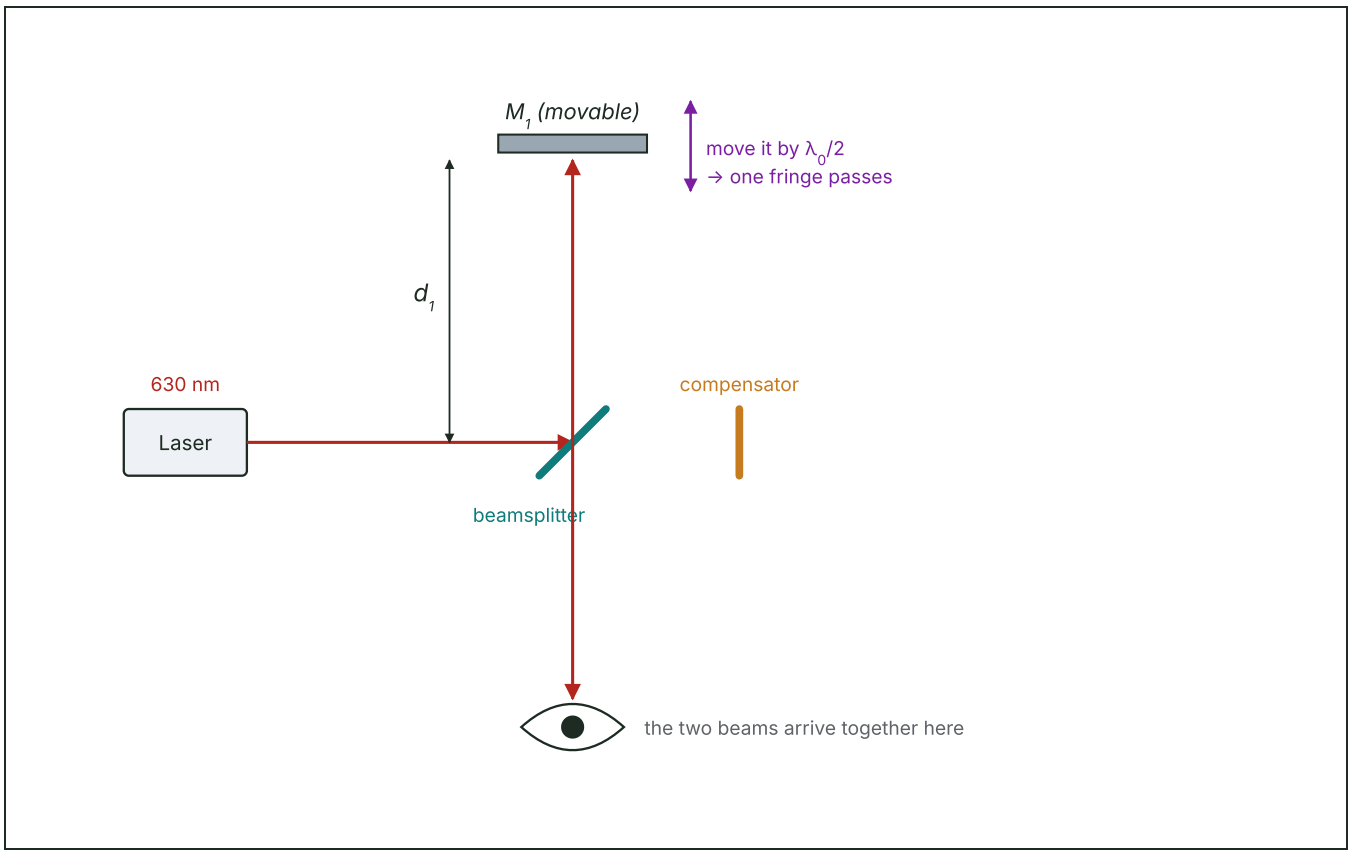


Figure 1. The beam is split in two, the halves travel down **separate arms**, and they are recombined. The compensator plate puts the same thickness of glass in both arms so that the only difference left is the one you control. **Move M_1 and the path difference changes — by twice the distance you moved it**, because the light goes down the arm and back.

Your job on Figure 1. The first arm is drawn: the beam goes up to M_1, a distance d_1 away. Draw the **second arm** — the beam through the compensator to the fixed mirror — and mark d_2. Then write down the total path difference.

The two half-beams travel $2d_1$ and $2d_2$ — down each arm and back — so the path difference is

$$\Delta d = 2d_1 - 2d_2$$

- $\Delta d = m\lambda_0 \rightarrow$ **constructive**
- $\Delta d = (m + ½)\lambda_0 \rightarrow$ **destructive**

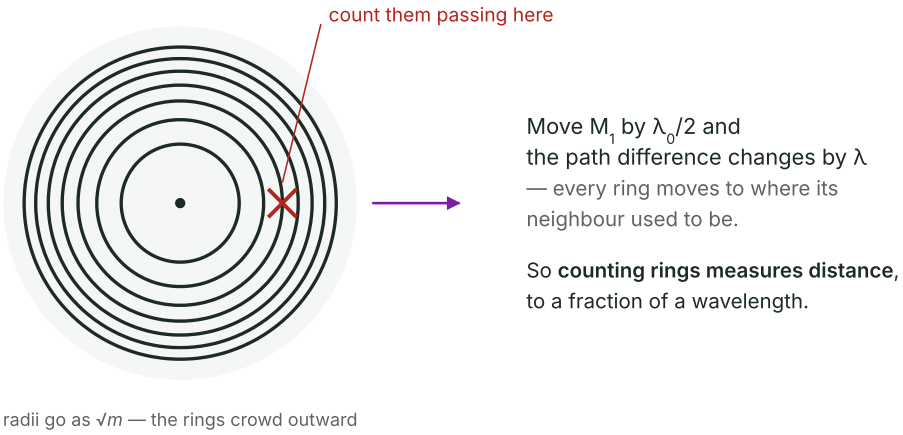


Figure 2. The fringes are circles, and they are **not evenly spaced** — the bright rings sit at radii proportional to √*m*, so they bunch up toward the edge. Pick one point, mark it, and count rings as they sweep past.

When M_1 is moved a distance $\Delta d = \lambda_0/2$, **the path difference changes by $\boldsymbol{\lambda_0}$** — the factor of two from the round trip — and each fringe moves to the position previously occupied by an adjacent fringe.

THE CONSEQUENCE

By counting the number of fringes *m* passing a single point,
we can measure minute displacement accurate to a fraction of a wavelength

Worked example 1 — how far did the mirror move?

λ = **630 nm**. M_1 is moved and **a single fringe shifts**. How much did the mirror move?

Worked example 2 — the refractive index of a gas

A cell of length t = **2 cm** is placed in one arm and slowly filled with gas. With λ = **632.8 nm**, **122 fringes** shift as it fills. What is the refractive index of the gas?

1. Nothing moved. So what changed the path difference? Answer that first and the algebra is two lines.

THE PUNCHLINE

$$\text{Move the mirror } \lambda_0/2 \rightarrow \text{ one fringe passes}$$

Everything else about this instrument follows from the factor of two. A displacement too small to see becomes a count you can make by eye, and the same count — with nothing moving at all — weighs the optical thickness of a gas.

Looking ahead

Notes 10 and 11 treated interference and diffraction as separate effects. They are not: real slits have width *and* separation. **Note 14** puts the two patterns together, and finds that some of Note 10's bright fringes simply are not there.

2. With λ = 500 nm, a mirror is moved and **8** fringes pass. How far did it move, in nm?

_____ nm

3. In a Michelson, the circular fringes are spaced:

○ Evenly, like the fringes of a double slit. ○ With radii going as $\sqrt{m}$, so they crowd together outward. ○ With radii going as m^2, so they spread out. ○ Randomly; the spacing carries no information.

4. A 5.0 cm cell is filled with a gas of n = 1.00050 using λ = 500 nm. How many fringes shift?

_____

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.